Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a\frac{r^n-1}{r-1}$$

Infinite sum

$$S_\infty = \frac{a}{1-r} \ (|r| < 1)$$

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Worked Solutions **Geometric Sequences**

Q001 Written

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Solution

1. Here $a = 2$ and $r = 3$.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$2 \times 3^{n-1}$

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Solution

1. $a_n = 3 \times 4^{n-1}$

Final answer

$3 \times 4^{n-1}$

Worked Solutions **Geometric Sequences**

Q003 **Written**

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5 \times 2^{n-1}$.

Final answer

$5 \times 2^{n-1}$

Q004 **Written**

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2 \times 5^{n-1}$.

Final answer

$2 \times 5^{n-1}$

Worked Solutions **Geometric Sequences**

Q005 Written

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $7 \times 4^{n-1}$.

Final answer

$7 \times 4^{n-1}$

Q006 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = 3$.

Correct option: A

Final answer

$a_n = 2\,(3)^4$

Worked Solutions **Geometric Sequences**

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = 2$.

Correct option: A

Final answer

$a_n = 3\,(2)^5$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 4$.

Correct option: A

Final answer

$a_n = 5\,(4)^3$

Worked Solutions **Geometric Sequences**

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = 5$.

Correct option: A

Final answer

$a_n = 4\,(5)^2$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 2$.

Correct option: A

Final answer

$a_n = 5\,(2)^6$

Worked Solutions **Geometric Sequences**

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 7$ and $r = 4$.

Correct option: A

Final answer

$a_n = 7\,(4)^2$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2.

Solution

1. $a_n = 5(-2)^{n-1}$

Final answer

$5(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q013 **Written**

Find the general term of the geometric sequence in part d. The first term is 2, common ratio −3.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$2(-3)^{n-1}$

Q014 **Written**

Find the general term of the geometric sequence in part e. The first term is 3, common ratio −5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3(-5)^{n-1}$.

Final answer

$3(-5)^{n-1}$

Worked Solutions **Geometric Sequences**

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4(-2)^{n-1}$.

Final answer

$4(-2)^{n-1}$

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = -3$.

Correct option: A

Final answer

$a_n = 2\,(-3)^4$

Worked Solutions **Geometric Sequences**

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -2$.

Correct option: A

Final answer

$a_n = 3(-2)^5$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = -1$.

Correct option: A

Final answer

$a_n = 6(-1)^3$

Worked Solutions **Geometric Sequences**

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -2$.

Correct option: A

Final answer

$a_n = 8\,(-2)^2$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 9$ and $r = -2$.

Correct option: A

Final answer

$a_n = 9\,(-2)^4$

Worked Solutions **Geometric Sequences**

Q021 **Written**

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Solution

1. $a_n = 8\left(-\frac{1}{3}\right)^{n-1}$

Final answer

$8\left(-\frac{1}{3}\right)^{n-1}$

Q022 **Written**

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4\left(\frac{1}{3}\right)^{n-1}$.

Final answer

$4\left(\frac{1}{3}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{1}{2}\right)^{n-1}$.

Final answer

$3\left(-\frac{1}{2}\right)^{n-1}$

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $6\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$6\left(\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = \frac{1}{3}$.

Correct option: A

Final answer

$a_n = 4\left(\frac{1}{3}\right)^4$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -\frac{1}{2}$.

Correct option: A

Final answer

$a_n = 3\left(-\frac{1}{2}\right)^5$

Worked Solutions **Geometric Sequences**

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = \frac{1}{2}$.

Correct option: A

Final answer

$a_n = 6\left(\frac{1}{2}\right)^3$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = \frac{3}{2}$.

Correct option: A

Final answer

$a_n = 2\left(\frac{3}{2}\right)^4$

Worked Solutions **Geometric Sequences**

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -\frac{3}{2}$.

Correct option: A

Final answer

$a_n = 8\left(-\frac{3}{2}\right)^3$

Q030 Written

Find the general term of $5, 10, 20, 40, \ldots$.

Solution

1. The common ratio is $r = 10/5 = 2$.
2. $a_n = 5 \times 2^{n-1}$

Final answer

$5 \times 2^{n-1}$

Worked Solutions **Geometric Sequences**

Q031 **Written**

Find the general term of $2, 6, 18, 54, \ldots$.

Solution

1. The common ratio is 3.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$2 \times 3^{n-1}$

Q032 **Written**

Find the general term of the geometric sequence in part j. The sequence is $3, 6, 12, 24, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3 \times 2^{n-1}$.

Final answer

$3 \times 2^{n-1}$

Worked Solutions **Geometric Sequences**

Q033 *MCQ*

Select the correct option: Find the general term of the geometric sequence $3, 6, 12, 24, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$3 \times 2^{n-1}$

Q034 *MCQ*

Select the correct option: Find the general term of the geometric sequence $2, 10, 50, 250, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$2 \times 5^{n-1}$

Worked Solutions **Geometric Sequences**

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence $4, 8, 16, 32, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$4 \times 2^{n-1}$

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence $7, 21, 63, 189, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$7 \times 3^{n-1}$

Worked Solutions **Geometric Sequences**

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence $1, 4, 16, 64, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

4^{n-1}

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is $2, -6, 18, -54, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.

2. The result is $2(-3)^{n-1}$.

Final answer

$2(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is $5, -10, 20, -40, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5(-2)^{n-1}$.

Final answer

$5(-2)^{n-1}$

Q040 MCQ

Select the correct option: Find the general term of the geometric sequence $5, -10, 20, -40, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$5(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q041 *MCQ*

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$2(-3)^{n-1}$

Q042 *MCQ*

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$-4(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence $6, -18, 54, -162, \ldots$**:**

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$6(-3)^{n-1}$

Q044 Written

Find the general term of $-4, 6, -9, \frac{27}{2}, \ldots$**.**

Solution

1. The common ratio is $r = 6/(-4) = -3/2$.
2. $a_n = -4\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$-4\left(-\frac{3}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q045 **Written**

Find the general term of $-8, 12, -18, 27, \ldots.$

Solution

1. The common ratio is $-3/2$.
2. $a_n = -8\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 **Written**

Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots.$

Solution

1. The common ratio is $1/3$.
2. $a_n = 2\left(\frac{1}{3}\right)^{n-1}$

Final answer

$2\left(\frac{1}{3}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q047 Written

Find the general term of the geometric sequence in part m. The sequence is $48, 24, 12, 6, \ldots.$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $48\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \ldots.$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$.

Final answer

$$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$$

Worked Solutions **Geometric Sequences**

Q049 **Written**

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $8\left(2\sqrt{2}\right)^{n-1}$.

Final answer

$8\left(2\sqrt{2}\right)^{n-1}$

Q050 **Written**

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{5}{2}\right)^{n-1}$.

Final answer

$3\left(-\frac{5}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$.

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5\left(-\frac{2}{3}\right)^{n-1}$.

Final answer

$$5\left(-\frac{2}{3}\right)^{n-1}$$

Worked Solutions **Geometric Sequences**

Q053 *MCQ*

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$-8\left(-\frac{3}{2}\right)^{n-1}$

Q054 *MCQ*

Select the correct option: Find the general term of the geometric sequence $48, 24, 12, 6, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$48\left(\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \ldots$**:**

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \ldots$**:**

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$8\left(2\sqrt{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q057 **MCQ**

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q058 **MCQ**

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$3\left(-\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$10\left(\frac{1}{2}\right)^{n-1}$

Q060 Written

Find the common ratio and missing values in $A, -18, 6, B, \ldots$.

Solution

1. $r = 6/(-18) = -\frac{1}{3}$

2. $A = 54, \quad B = 6\left(-\frac{1}{3}\right) = -2$

Final answer

$r = -\frac{1}{3},\ A = 54,\ B = -2$

Worked Solutions **Geometric Sequences**

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \ldots$ is geometric. Find r, A, B:

Solution

1. Divide adjacent known terms to find r, then extend one step in each direction.

Correct option: A

Final answer

$r = \frac{1}{2},\ A = 24,\ B = 3$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \ldots$.

Solution

1. $24r = 12 \Rightarrow r = \frac{1}{2}$
2. $A\left(\frac{1}{2}\right) = 24 \Rightarrow A = 48, \quad B = 12\left(\frac{1}{2}\right) = 6$

Final answer

$r = \frac{1}{2},\ A = 48,\ B = 6$

Worked Solutions **Geometric Sequences**

Q063 **Written**

Find the common ratio and missing values in $A, -3, 9, B$.

Solution

1. $r = -3, \quad A = 1, \quad B = -27$

Final answer

$r = -3,\ A = 1,\ B = -27$

Q064 **Written**

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

Solution

1. $r = \frac{1}{3}, \quad A = 1, \quad B = \frac{1}{27}$

Final answer

$r = \dfrac{1}{3},\ A = 1,\ B = \dfrac{1}{27}$

Worked Solutions **Geometric Sequences**

Q065 **Written**

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Solution

1. $r = -\frac{1}{2}, \quad A = \frac{1}{2}, \quad B = -\frac{1}{16}$

Final answer

$r = -\frac{1}{2}$, $A = \frac{1}{2}$, $B = -\frac{1}{16}$

Worked Solutions **Geometric Sequences**

Q066 **Challenge**

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7.

Solution

1. $a_3 = \frac{24}{r}, \quad a_5 = 24r, \quad 24\left(r + \frac{1}{r}\right) = 80$
2. $3r^2 - 10r + 3 = 0 \Rightarrow r = 3 \text{ or } \frac{1}{3}$
3. The sequence is increasing, so $r = 3$; hence $a_7 = 24(3)^3 = 648$.

Final answer

$a_7 = 648$

Worked Solutions **Geometric Sequences**

Q067 **Written**

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Solution

1. $r^2 = 96/24 = 4 \Rightarrow r = \pm 2$
2. $(a, r) = (3, 2)$ or $(-3, -2)$
3. Therefore both real branches are retained.

Final answer

$a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 **Written**

Find the general term when the 3rd term is 18 and the 5th term is 162.

Solution

1. $r^2 = 162/18 = 9 \Rightarrow r = \pm 3$
2. $a = 18/r^2 = 2$

Final answer

$a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q069 **Written**

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Solution

1. $r^2 = 54/6 = 9 \Rightarrow r = \pm 3$
2. $a = 6/r$

Final answer

$a_n = 2(3)^{n-1} \text{ or } -2(-3)^{n-1}$

Q070 **Written**

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Solution

1. $r^2 = 243/27 = 9 \Rightarrow r = \pm 3$
2. $a = 3$

Final answer

$a_n = 3(3)^{n-1} \text{ or } 3(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q071 **Written**

Find the requested values. The 2nd term is 14 and the 5th term is 112.

Solution

1. $r^3 = 112/14 = 8 \Rightarrow r = 2$
2. $a = 7$

Final answer

$a_n = 7(2)^{n-1}$

Q072 **Written**

Find the general term when the 3rd term is 12 and the 6th term is −96.

Solution

1. $r^3 = -96/12 = -8 \Rightarrow r = -2$
2. $a = 12/r^2 = 3$
3. $a_n = 3(-2)^{n-1}$

Final answer

$a_n = 3(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48.

Solution

1. $r^3 = 48/(-6) = -8 \Rightarrow r = -2$
2. $a = -6/r = 3$

Final answer

$a_n = 3(-2)^{n-1}$

Q074 Written

Find the general term when the 3rd term is 12 and the 7th term is 192.

Solution

1. $r^4 = 192/12 = 16 \Rightarrow r = \pm 2$
2. $a = 12/r^2 = 3$

Final answer

$a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q075 Written

Find the requested values. The 3rd term is -28 and the 5th term is -112.

Solution

1. $r^2 = 4 \Rightarrow r = \pm 2$
2. $a = -28/4 = -7$

Final answer

$a_n = -7(2)^{n-1} \text{ or } -7(-2)^{n-1}$

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128.

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -2$

Final answer

$a_n = -2(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q077 **Written**

Find the requested values. The 3rd term is -20 and the 6th term is 160.

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -5$

Final answer

$a_n = -5(-2)^{n-1}$

Q078 **Written**

Find the requested values. The 2nd term is 3 and the 6th term is 768.

Solution

1. $r^4 = 768/3 = 256 \Rightarrow r = \pm 4$
2. $a = 3/r$

Final answer

$a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Worked Solutions **Geometric Sequences**

Q079 Written

Find the requested values. The 3rd term is 9 and the 7th term is 729/16.

Solution

1. $r^4 = 81/16 \Rightarrow r = \pm\frac{3}{2}$
2. $a = 4$

Final answer

$a_n = 4(\frac{3}{2})^{n-1}$ or $4(-\frac{3}{2})^{n-1}$

Q080 Written

Find the requested values. The 4th term is 2/9 and the 8th term is 18.

Solution

1. $r^4 = 81 \Rightarrow r = \pm 3$
2. $a = \frac{2/9}{r^3} = \pm\frac{2}{243}$

Final answer

$a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 2\,(3)^{n-1}$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 3\,(2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 5\,(4)^{n-1}$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 4\,(5)^{n-1}$

Worked Solutions

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 2\,(-3)^{n-1}$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 3\,(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q087 *MCQ*

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 6\,(-1)^{n-1}$

Q088 *MCQ*

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 8\,(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q089 MCQ

Select the correct option: For $a = 4, r = \frac{1}{3}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 4\left(\frac{1}{3}\right)^{n-1}$

Q090 MCQ

Select the correct option: For $a = 3, r = -\frac{1}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 3\left(-\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q091 MCQ

Select the correct option: For $a = 6, r = \frac{1}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 6\left(\frac{1}{2}\right)^{n-1}$

Q092 MCQ

Select the correct option: For $a = 2, r = \frac{3}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 2\left(\frac{3}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q093 MCQ

Select the correct option: For $a = 8, r = -\frac{3}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 8\left(-\frac{3}{2}\right)^{n-1}$

Q094 MCQ

Select the correct option: For $a = 5, r = 2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 5\,(2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q095 MCQ

Select the correct option: For $a = 7$, $r = 4$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 7\,(4)^{n-1}$

Q096 MCQ

Select the correct option: For $a = 9$, $r = -2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 9\,(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q097 Written

When $2, x, 8$ form a geometric sequence in this order, find x.

Solution

1. $x^2 = 2 \times 8 = 16$
2. $x = \pm 4$

Final answer

$x = \pm 4$

Q098 Written

Find the requested values. When $1, x, 25$ form a geometric sequence, find x.

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions **Geometric Sequences**

Q099 **Written**

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x.

Solution

1. $x^2 = \frac{1}{2} \times 8 = 4 \Rightarrow x = \pm 2$

Final answer

$x = \pm 2$

Q100 **Written**

When $1, x, 12$ form a geometric sequence in this order, find x.

Solution

1. $x^2 = 1 \times 12 = 12$
2. $x = \pm 2\sqrt{3}$

Final answer

$x = \pm 2\sqrt{3}$

Worked Solutions **Geometric Sequences**

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x:

Solution

1. Use $x^2 = (2)(8)$.

Correct option: A

Final answer

± 4

Q102 MCQ

Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x:

Solution

1. Use $x^2 = (1)(25)$.

Correct option: A

Final answer

± 5

Worked Solutions **Geometric Sequences**

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x:

Solution

1. Use $x^2 = (\frac{1}{2})(8)$.

Correct option: A

Final answer

± 2

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple $3, x, 12$, find x:

Solution

1. Use $x^2 = (3)(12)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple $4, x, 9$, find x:

Solution

1. Use $x^2 = (4)(9)$.

Correct option: A

Final answer

± 6

Q106 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 18$, find x:

Solution

1. Use $x^2 = (2)(18)$.

Correct option: A

Final answer

± 6

Worked Solutions **Geometric Sequences**

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b.

Solution

1. $2a = 6 + b \Rightarrow b = 2a - 6$
2. $b^2 = 16a \Rightarrow 4a^2 - 40a + 36 = 0$
3. $(a, b) = (9, 12)$ or $(1, -4)$

Final answer

$(a, b) = (9, 12)$ or $(1, -4)$

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b.

Solution

1. $2b = a + 2$
2. $b^2 = 2a$
3. $(a, b) = (2, 2)$ or $(-4, -1)$

Final answer

$(a, b) = (2, 2)$ or $(-4, -1)$

Worked Solutions **Geometric Sequences**

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b.

Solution

1. $b = 2a + 2$
2. $b^2 = 18a \Rightarrow (a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$

Final answer

$(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$

Worked Solutions **Geometric Sequences**

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b.

Solution

1. $b = 2a + 5$
2. $b^2 = 45a \Rightarrow 4a^2 - 25a + 25 = 0$
3. $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Final answer

$(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Worked Solutions **Geometric Sequences**

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Solution

1. Use $S_5 = \dfrac{4(3^5-1)}{3-1} = 2(3^5-1)$.
2. Therefore $S_5 = 484$.

Final answer

484

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Solution

1. $S_5 = \dfrac{3(2^5-1)}{2-1} = 3(31)$.
2. Therefore $S_5 = 93$.

Final answer

93

Worked Solutions **Geometric Sequences**

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Solution

1. $S_4 = \dfrac{5(3^4 - 1)}{3 - 1} = 5(40)$.
2. Therefore $S_4 = 200$.

Final answer

200

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Solution

1. $S_{10} = \dfrac{3(2^{10} - 1)}{2 - 1} = 3(1023)$.
2. Therefore $S_{10} = 3069$.

Final answer

3069

Worked Solutions — **Geometric Sequences**

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{4(1-(3)^4)}{1-(3)}$.
2. Therefore $S_4 = 160$.

Correct option: D

Final answer

160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{5(1-(2)^6)}{1-(2)}$.
2. Therefore $S_6 = 315$.

Correct option: C

Final answer

315

Worked Solutions **Geometric Sequences**

Q117 *MCQ*

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{2(1-(4)^5)}{1-(4)}$.
2. Therefore $S_5 = 682$.

Correct option: B

Final answer

682

Q118 *MCQ*

For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{3(1-(5)^4)}{1-(5)}$.
2. Therefore $S_4 = 468$.

Correct option: A

Final answer

468

Worked Solutions **Geometric Sequences**

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{7(1-(2)^5)}{1-(2)}$.
2. Therefore $S_5 = 217$.

Correct option: D

Final answer

217

Q120 MCQ

For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{6(1-(3)^3)}{1-(3)}$.
2. Therefore $S_3 = 78$.

Correct option: C

Final answer

78

Worked Solutions **Geometric Sequences**

Q121 **Written**

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. Use $S_6 = \dfrac{8(1-(1/2)^6)}{1-1/2} = 16\{1-(1/2)^6\}$.
2. Therefore $S_6 = \dfrac{63}{4}$.

Final answer

$\dfrac{63}{4}$

Q122 **Written**

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. $S_6 = \dfrac{6(1-(1/2)^6)}{1-1/2} = 12\{1-(1/2)^6\}$.
2. Therefore $S_6 = \dfrac{189}{16}$.

Final answer

$\dfrac{189}{16}$

Worked Solutions **Geometric Sequences**

Q123 **Written**

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Solution

1. $S_3 = \dfrac{1-(1/3)^3}{1-1/3} = \dfrac{26}{27} \cdot \dfrac{3}{2}$.
2. Therefore $S_3 = \dfrac{13}{9}$.

Final answer

$\dfrac{13}{9}$

Q124 **Written**

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Solution

1. $S_8 = \dfrac{1-(1/2)^8}{1-1/2} = 2\{1-(1/2)^8\}$.
2. Therefore $S_8 = \dfrac{255}{128}$.

Final answer

$\dfrac{255}{128}$

Worked Solutions **Geometric Sequences**

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{8(1 - (\frac{1}{2})^4)}{1 - (\frac{1}{2})}$.
2. Therefore $S_4 = 15$.

Correct option: B

Final answer

15

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{9(1 - (\frac{1}{3})^3)}{1 - (\frac{1}{3})}$.
2. Therefore $S_3 = 13$.

Correct option: A

Final answer

13

Worked Solutions **Geometric Sequences**

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{12(1 - (\frac{1}{2})^5)}{1 - (\frac{1}{2})}$.
2. Therefore $S_5 = \dfrac{93}{4}$.

Correct option: D

Final answer

$\dfrac{93}{4}$

Q128 **MCQ**

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{5(1 - (\frac{1}{4})^3)}{1 - (\frac{1}{4})}$.
2. Therefore $S_3 = \dfrac{105}{16}$.

Correct option: C

Final answer

$\dfrac{105}{16}$

Worked Solutions **Geometric Sequences**

Q129 **MCQ**

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{10(1 - (\frac{1}{2})^6)}{1 - (\frac{1}{2})}$.
2. Therefore $S_6 = \dfrac{315}{16}$.

Correct option: B

Final answer
$\frac{315}{16}$

Q130 **MCQ**

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{4(1 - (\frac{1}{3})^4)}{1 - (\frac{1}{3})}$.
2. Therefore $S_4 = \dfrac{160}{27}$.

Correct option: A

Final answer

$\dfrac{160}{27}$

Worked Solutions **Geometric Sequences**

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2, and 4 terms.

Solution

1. $S_4 = \dfrac{6\{1-(-2)^4\}}{1-(-2)} = \dfrac{6(-15)}{3}$.
2. Therefore $S_4 = -30$.

Final answer

-30

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3, and 4 terms.

Solution

1. $S_4 = \dfrac{1-(-3)^4}{1-(-3)} = \dfrac{-80}{4}$.
2. Therefore $S_4 = -20$.

Final answer

-20

Worked Solutions **Geometric Sequences**

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3, and 6 terms.

Solution

1. $S_6 = \dfrac{2\{1-(-3)^6\}}{1-(-3)} = \dfrac{2(-728)}{4}$.
2. Therefore $S_6 = -364$.

Final answer

-364

Q134 MCQ

For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{4(1-(-2)^5)}{1-(-2)}$.
2. Therefore $S_5 = 44$.

Correct option: D

Final answer

44

Worked Solutions **Geometric Sequences**

Q135 **MCQ**

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{3(1-(-3)^4)}{1-(-3)}$.
2. Therefore $S_4 = -60$.

Correct option: C

Final answer

-60

Q136 **MCQ**

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{2(1-(-1)^6)}{1-(-1)}$.
2. Therefore $S_6 = 0$.

Correct option: B

Final answer

0

Worked Solutions **Geometric Sequences**

Q137 *MCQ*

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{5(1-(-2)^6)}{1-(-2)}$.
2. Therefore $S_6 = -105$.

Correct option: A

Final answer

-105

Q138 *MCQ*

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{7(1-(-1)^5)}{1-(-1)}$.
2. Therefore $S_5 = 7$.

Correct option: D

Final answer

7

Worked Solutions   **Geometric Sequences**

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{6(1-(-3)^3)}{1-(-3)}$.
2. Therefore $S_3 = 42$.

Correct option: C

Final answer

42

Q140 Written

Find S_n from the first term to the n-th term when the first term is 3 and the common ratio is -2.

Solution

1. Here $a = 3$ and $r = -2$.
2. $S_n = \dfrac{3\{1-(-2)^n\}}{1-(-2)} = 1-(-2)^n$.

Final answer

$1-(-2)^n$

Worked Solutions **Geometric Sequences**

Q141 Written

Find S_n from the first term to the n-th term when the first term is 7 and the common ratio is 2.

Solution

1. $S_n = \dfrac{7(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 7(2^n - 1)$.

Final answer

$7(2^n - 1)$

Q142 Written

Find S_n from the first term to the n-th term when the first term is 2 and the common ratio is $\dfrac{1}{3}$.

Solution

1. $S_n = \dfrac{2\{1 - (1/3)^n\}}{1 - 1/3}$.
2. Hence $S_n = 3\{1 - (1/3)^n\}$.

Final answer

$3\{1 - (\frac{1}{3})^n\}$

Worked Solutions **Geometric Sequences**

Q143 Written

Find S_n from the first term to the n-th term when the first term is 4 and the common ratio is 2.

Solution

1. $S_n = \dfrac{4(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 4(2^n - 1)$.

Final answer

$4(2^n - 1)$

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n-th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \dfrac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 3(2^n - 1)$.

Correct option: C

Final answer

$3(2^n - 1)$

Worked Solutions **Geometric Sequences**

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n-th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \dfrac{a(1-r^n)}{1-r}$ to obtain $S_n = 10\{1 - (\frac{1}{2})^n\}$.

Correct option: D

Final answer

$10\{1 - (\frac{1}{2})^n\}$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n-th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \dfrac{a(1-r^n)}{1-r}$ to obtain $S_n = \dfrac{4\{1-(-2)^n\}}{3}$.

Correct option: A

Final answer

$$\frac{4\{1-(-2)^n\}}{3}$$

Worked Solutions **Geometric Sequences**

Q147 **Written**

Find S_n from the first term to the n-th term of $2, 6, 18, 54, \ldots$.

Solution

1. The sequence has $a = 2$ and $r = 3$.
2. $S_n = \dfrac{2(3^n - 1)}{3 - 1} = 3^n - 1$.

Final answer

$3^n - 1$

Q148 **Written**

Find S_n from the first term to the n-th term of $2, -6, 18, -54, \ldots$.

Solution

1. The sequence has $a = 2$ and $r = -3$.
2. $S_n = \dfrac{2\{1 - (-3)^n\}}{1 - (-3)} = \dfrac{1 - (-3)^n}{2}$.

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions **Geometric Sequences**

Q149 Written

Find S_n from the first term to the n-th term of $8, -4, 2, -1, \ldots$.

Solution

1. The sequence has $a = 8$ and $r = -\frac{1}{2}$.
2. $S_n = \dfrac{8\{1-(-1/2)^n\}}{1+1/2} = \dfrac{16}{3}\{1-(-1/2)^n\}$.

Final answer

$\frac{16}{3}\{1-(-\frac{1}{2})^n\}$

Q150 Written

Find S_n from the first term to the n-th term of $1, -3, 9, -27, \ldots$.

Solution

1. The sequence has $a = 1$ and $r = -3$.
2. $S_n = \dfrac{1-(-3)^n}{1+3} = \dfrac{1-(-3)^n}{4}$.

Final answer

$\dfrac{1-(-3)^n}{4}$

Worked Solutions **Geometric Sequences**

Q151 **Written**

Find S_n from the first term to the n-th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Solution

1. $S_n = \dfrac{16\{1-(-1/2)^n\}}{1+1/2}$.
2. Hence $S_n = \dfrac{32}{3}\{1-(-1/2)^n\}$.

Final answer

$\frac{32}{3}\{1-(-\frac{1}{2})^n\}$

Q152 **Written**

Find S_n from the first term to the n-th term of $1, -2, 4, -8, \ldots$.

Solution

1. The sequence has $a = 1$ and $r = -2$.
2. $S_n = \dfrac{1-(-2)^n}{1+2} = \dfrac{1-(-2)^n}{3}$.

Final answer

$\dfrac{1-(-2)^n}{3}$

Worked Solutions **Geometric Sequences**

Q153 MCQ

A culture grows by $3, 9, 27, 81$ **units. Find the four-stage total:**

Solution

1. Recognise the listed stages as a finite geometric sequence and identify its first term and ratio.
2. Apply the finite-GP formula to obtain $S_n = 120$.

Correct option: A

Final answer

120

Worked Solutions **Geometric Sequences**

Q154 **Challenge**

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \ldots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Solution

1. Here $a = 3$, $r = \frac{1}{3}$, and $n = 4$.
2. $S_4 = \dfrac{3\{1 - (1/3)^4\}}{1 - 1/3} = \dfrac{40}{9}$ mg/L.

Final answer

$\frac{40}{9}$

Worked Solutions **Geometric Sequences**

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r):

Solution

1. The shifted block is $r^2S_3 = 28$, so $r^2 = 4$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 1$ or $a = \frac{7}{3}$, respectively.

Correct option: D

Final answer

$(a, r) = (1, 2)$ or $(a, r) = \left(\frac{7}{3}, -2\right)$

Worked Solutions **Geometric Sequences**

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 234$, so $r^2 = 9$; then $a(1 + r + r^2) = 26$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 2$ or $a = \frac{26}{7}$, respectively.

Correct option: C

Final answer

$(a, r) = (2, 3)$ or $(a, r) = \left(\frac{26}{7}, -3\right)$

Worked Solutions **Geometric Sequences**

Q157 *MCQ*

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$; then $a(1 + r + r^2) = 21$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 3$ or $a = 7$, respectively.

Correct option: B

Final answer

$(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

Worked Solutions **Geometric Sequences**

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 112$, so $r^2 = 4$; then $a(1 + r + r^2) = 28$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: A

Final answer

$(a, r) = (4, 2)$ or $(a, r) = \left(\frac{28}{3}, -2\right)$

Worked Solutions **Geometric Sequences**

Q159 *MCQ*

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 117$, so $r^2 = 9$; then $a(1 + r + r^2) = 13$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 1$ or $a = \dfrac{13}{7}$, respectively.

Correct option: D

Final answer

$(a, r) = (1, 3)$ or $(a, r) = \left(\dfrac{13}{7}, -3\right)$

Worked Solutions **Geometric Sequences**

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$; then $a(1 + r + r^2) = 35$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 5$ or $a = \frac{35}{3}$, respectively.

Correct option: C

Final answer

$(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

Worked Solutions **Geometric Sequences**

Q161 MCQ

Given $S_3 = \dfrac{19}{2}$ and the sum from the 3rd to the 5th terms is $\dfrac{171}{8}$, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = \dfrac{171}{8}$, so $r^2 = \dfrac{9}{4}$; then $a(1 + r + r^2) = \dfrac{19}{2}$.
2. Solving in the real numbers gives $r = \dfrac{3}{2}$ or $r = -\dfrac{3}{2}$.
3. Back-substitution gives $a = 2$ or $a = \dfrac{38}{7}$, respectively.

Correct option: D

Final answer

$(a, r) = \left(2, \dfrac{3}{2}\right)$ or $(a, r) = \left(\dfrac{38}{7}, -\dfrac{3}{2}\right)$

Worked Solutions **Geometric Sequences**

Q162 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72.

Solution

1. The 4th+6th block is $r^3 S_3 = -72$, so $r^3 = -8$ and $r = -2$.
2. Then $a(1 - 2 + 4) = 9$, giving $a = 3$.

Final answer

$(a, r) = (3, -2)$

Worked Solutions **Geometric Sequences**

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = \frac{7}{4}$, so $r^2 = \frac{1}{4}$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = -\frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: C

Final answer

$(a, r) = \left(4, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q164 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Solution

1. Write $S_3 = a(1 + r + r^2) = 35$. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$ and $r = \pm 2$.
2. For $r = 2$, $7a = 35$, giving $a = 5$. For $r = -2$, $3a = 35$, giving $a = \frac{35}{3}$.

Final answer

$(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$ and $r = \pm 2$.
2. Using $a(1 + r + r^2) = 21$ gives $a = 3$ for $r = 2$ and $a = 7$ for $r = -2$.

Final answer

$(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

Worked Solutions **Geometric Sequences**

Q166 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Solution

1. The shifted block gives $r^2 S_3 = 36$, hence $r = \pm 2$.
2. Substitution into $a(1 + r + r^2) = 9$ gives $a = \dfrac{9}{7}$ for $r = 2$, or $a = 3$ for $r = -2$.

Final answer

$(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Solution

1. The shifted block gives $r^2 S_3 = 702$, so $r^2 = 9$ and $r = \pm 3$.
2. Using $a(1 + r + r^2) = 78$ gives $a = 6$ for $r = 3$, or $a = \dfrac{78}{7}$ for $r = -3$.

Final answer

$(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$

Worked Solutions **Geometric Sequences**

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 24$, so $r^2 = 4$; then $a(1 + r + r^2) = 6$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 2$ or $a = \frac{6}{7}$, respectively.

Correct option: B

Final answer

$(a, r) = (2, -2)$ or $(a, r) = \left(\frac{6}{7}, 2\right)$

Worked Solutions **Geometric Sequences**

Q169 **MCQ**

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 36$, so $r^2 = 4$; then $a(1 + r + r^2) = 9$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 3$ or $a = \frac{9}{7}$, respectively.

Correct option: A

Final answer

$(a, r) = (3, -2)$ or $(a, r) = \left(\frac{9}{7}, 2\right)$

Worked Solutions **Geometric Sequences**

Q170 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Solution

1. From $ar = 3$, write $a = 3/r$. Then $3/r + 3 + 3r = 13$, so $3r^2 - 10r + 3 = 0$.
2. Hence $r = 3$ or $r = \frac{1}{3}$, giving $a = 1$ or $a = 9$, respectively.

Final answer

$(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Solution

1. From $ar = 6$, $a = 6/r$. Substitution into $S_3 = 21$ gives $6/r + 6 + 6r = 21$, hence $2r^2 - 5r + 2 = 0$.
2. Therefore $r = 2$ or $r = \frac{1}{2}$, giving $a = 3$ or $a = 12$, respectively.

Final answer

$(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q172 **MCQ**

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r):

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 21$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 3$ or $a = 12$, respectively.

Correct option: C

Final answer

$(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r):

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 26$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 2$ or $a = 18$, respectively.

Correct option: D

Final answer

$(a, r) = (2, 3)$ or $(a, r) = \left(18, \frac{1}{3}\right)$

Worked Solutions **Geometric Sequences**

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r):

Solution

1. Use $ar = 8$ and $a(1 + r + r^2) = 28$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = 16$, respectively.

Correct option: B

Final answer

$(a, r) = (4, 2)$ or $(a, r) = \left(16, \frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r):

Solution

1. Use $ar = 3$ and $a(1 + r + r^2) = 13$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Correct option: C

Final answer

$(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

Worked Solutions **Geometric Sequences**

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r):

Solution

1. Use $ar = \frac{5}{2}$ and $a(1 + r + r^2) = \frac{35}{4}$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = 2$.
3. Back-substitution gives $a = 5$ or $a = \frac{5}{4}$, respectively.

Correct option: D

Final answer

$(a, r) = \left(5, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{5}{4}, 2\right)$

Worked Solutions **Geometric Sequences**

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Solution

1. The data give $ar^2 = 4$ and $a + ar + ar^2 = 7$. Substitute $a = 4/r^2$: $4 + 4r + 4r^2 = 7r^2$, so $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Final answer

$(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Solution

1. Use $ar^2 = 12$ and $a + ar + ar^2 = 9$. After multiplying by r^2, $r^2 + 4r + 4 = 0$, so $r = -2$.
2. Then $a = 12/(-2)^2 = 3$.

Final answer

$(a, r) = (3, -2)$

Worked Solutions **Geometric Sequences**

Q179 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Solution

1. Use $ar^2 = 20$ and $a + ar + ar^2 = 35$. Multiplication by r^2 gives $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$, and $a = 5$ or $a = 45$, respectively.

Final answer

$(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Solution

1. Use $S_6 = S_3 + r^3 S_3$: $189 = 21(1 + r^3)$, hence $r^3 = 8$ and $r = 2$ in the real numbers.
2. Then $a(1 + 2 + 4) = 21$, so $a = 3$.

Final answer

$(a, r) = (3, 2)$

Worked Solutions **Geometric Sequences**

Q181 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Solution

1. Since $S_6 = S_3 + r^3 S_3$, $728 = 26(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $a(1 + 3 + 9) = 26$, giving $a = 2$.

Final answer

$(a, r) = (2, 3)$

Q182 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Solution

1. Use $S_6 = S_3(1 + r^3)$: $1820 = 65(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $13a = 65$, giving $a = 5$.

Final answer

$(a, r) = (5, 3)$

Worked Solutions **Geometric Sequences**

Q183 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Solution

1. Let $a + ar = 3$. Factor $S_4 = (a + ar)(1 + r^2)$, so $3(1 + r^2) = 51$ and $r = \pm 4$.
2. Then $a = 3/(1 + r)$, giving $a = \dfrac{3}{5}$ for $r = 4$, or $a = -1$ for $r = -4$.

Final answer

$(a, r) = \left(\dfrac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8, and $S_4 = -80$.

Solution

1. Factor $S_4 = (a + ar)(1 + r^2)$. Thus $-8(1 + r^2) = -80$, so $r = \pm 3$.
2. Using $a(1 + r) = -8$ gives $a = -2$ for $r = 3$, or $a = 4$ for $r = -3$.

Final answer

$(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$

Worked Solutions **Geometric Sequences**

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10}:

Solution

1. $a_7 = S_7 - S_6 = 64$ and $a_8 = S_8 - S_7 = 128$.
2. Thus $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 512$.

Correct option: D

Final answer

512

Worked Solutions **Geometric Sequences**

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10}.

Solution

1. $S_8 - S_6 = a_7 + a_8 = 255 - 63 = 192$. Together with $a_8 - a_7 = 64$, this gives $a_7 = 64$ and $a_8 = 128$.
2. Hence $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 128(4) = 512$.

Final answer

$a_7 = 64,\ a_8 = 128,\ r = 2,\ a_{10} = 512$

Worked Solutions **Geometric Sequences**

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For $a = 3, r = 2$, find a_6:

Solution

1. $a_6 = 3(2)^5 = 96$

Correct option: B

Final answer

96

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $2, x, 8$ are consecutive geometric terms, find x:

Solution

1. $x^2 = 16 \Rightarrow x = \pm 4$

Correct option: C

Final answer

± 4

Worked Solutions **Geometric Sequences**

Quiz MCQ 3 **Concept Quiz · MCQ**

For $1, 3, 9, \ldots$, find S_n:

Solution

1. $a = 1, r = 3$, so $S_n = (3^n - 1)/2$.

Correct option: B

Final answer

$$\frac{3^n - 1}{2}$$

Quiz MCQ 4 **Concept Quiz · MCQ**

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10}:

Solution

1. $a_7 = S_7 - S_6 = 64$, $a_8 = S_8 - S_7 = 128$, so $r = 2$.

2. Therefore $a_{10} = 128(2^2) = 512$.

Correct option: D

Final answer

512

Worked Solutions **Geometric Sequences**

Quiz WRITTEN 5 Concept Quiz · Written

Find the general term of $-4, 8, -16, 32, \ldots$.

Solution

1. The ratio is -2.
2. $a_n = -4(-2)^{n-1}$

Final answer

$-4(-2)^{n-1}$

Quiz WRITTEN 6 Concept Quiz · Written

If $1, x, 25$ are consecutive geometric terms, find x.

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions **Geometric Sequences**

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs $5, 15, 45, 135$. Find the total output.

Solution

1. $S_4 = 5(1+3+9+27) = 200$.

Final answer

200

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10}.

Solution

1. $a_7 = 64$, $a_8 = 128$, hence $r = 2$.
2. $a_{10} = a_8 r^2 = 512$.

Final answer

512

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8\left(2\sqrt{2}\right)^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3},\ A = 54,\ B = -2$

Q061 A

Q062 $r = \frac{1}{2},\ A = 48,\ B = 6$

Q063 $r = -3,\ A = 1,\ B = -27$

Q064 $r = \frac{1}{3},\ A = 1,\ B = \frac{1}{27}$

Q065 $r = -\frac{1}{2},\ A = \frac{1}{2},\ B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4(\frac{3}{2})^{n-1}$ or $4(-\frac{3}{2})^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A

Q096 A

Q097 $x = \pm 4$

Q098 $x = \pm 5$

Q099 $x = \pm 2$

Q100 $x = \pm 2\sqrt{3}$

Q101 A

Q102 A

Q103 A

Q104 A

Q105 A

Q106 A

Q107 $(a, b) = (9, 12)$ or $(1, -4)$

Q108 $(a, b) = (2, 2)$ or $(-4, -1)$

Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$

Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Q111 484

Q112 93

Q113 200

Q114 3069

Q115 D

Q116 C

Q117 B

Q118 A

Q119 D

Q120 C

Q121 $\frac{63}{4}$

Q122 $\frac{189}{16}$

Q123 $\frac{13}{9}$

Q124 $\frac{255}{128}$

Q125 B

Q126 A

Q127 D

Q128 C

Q129 B

Q130 A

Q131 -30

Q132 -20

Q133 -364

Q134 D

Q135 C

Q136 B

Q137 A

Q138 D

Q139 C

Q140 $1 - (-2)^n$

Q141 $7(2^n - 1)$

Q142 $3\{1 - (\frac{1}{3})^n\}$

Q143 $4(2^n - 1)$

Q144 C

Q145 D

Q146 A

Q147 $3^n - 1$

Q148 $\frac{1 - (-3)^n}{2}$

Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$

Q150 $\frac{1 - (-3)^n}{4}$

Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$

Q152 $\frac{1 - (-2)^n}{3}$

Q153 A

Q154 $\frac{40}{9}$

Q155 D

Q156 C

Q157 B

Q158 A

Q159 D

Q160 C

Q161 D

Q162 $(a, r) = (3, -2)$

Q163 C

Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$

Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$

Q168 B

Q169 A

Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

Q172 C

Q173 D

Q174 B

Q175 C

Q176 D

Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$

Q178 $(a, r) = (3, -2)$

Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$

Q180 $(a, r) = (3, 2)$

Q181 $(a, r) = (2, 3)$

Q182 $(a, r) = (5, 3)$

Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$

Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$

Q185 D

Q186 $a_7 = 64,\ a_8 = 128,\ r = 2,\ a_{10} = 512$

Quiz MCQ 1 B

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 D

Quiz WRITTEN 5 $-4(-2)^{n-1}$

Quiz WRITTEN 6 $x = \pm 5$

Quiz WRITTEN 7 200

Quiz WRITTEN 8 512